# Scaled-ActNet: Kolmogorov Superposition Networks with Frequency Scaling

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## Abstract

The Kolmogorov Superposition Theorem (KST) has inspired new neural architectures like Kolmogorov-Arnold Networks (KANs) and their variants, as well as ActNet, which have shown promise in machine learning applications. However, these models suffer from spectral bias, limiting their ability to represent high-frequency signals and thus making them unsuitable for implicit neural representation (INR) tasks. We introduce Scaled-ActNet, a novel architecture that bridges this gap by leveraging the frequency-aware principles of SIREN into a KST-based framework. Scaled-ActNet modifies the existing ActNet architecture by introducing an input-scaling frequency parameter, $\omega_0$, and a principled, SIREN-inspired initialization scheme of the learnable parameters that depends on it. We demonstrate the effectiveness of Scaled-ActNet on two domains: it outperforms ActNet by up to 23 dB in PSNR on 2D image reconstruction, and achieves a nearly 9x lower loss than SIREN on audio reconstruction. Our work establishes Scaled-ActNet as a versatile and powerful architecture for a range of signal representation tasks.

## 1 Introduction

Implicit Neural Representations (INRs) have become a dominant paradigm for representing complex, continuous signals such as images, audio, and video [1, 5]. Models such as SIREN (Sinusoidal Representation Networks) [1] have demonstrated a remarkable ability to capture high-frequency details by leveraging input scaling, sinusoidal activation functions and a carefully designed initialization scheme. Concurrently, the Kolmogorov Superposition Theorem (KST) has led to new architectures like Kolmogorov-Arnold Networks (KANs) [3] and ActNet [2], which show promise for scientific applications like solving Partial Differential Equations (PDEs) [4].

Despite their theoretical appeal, current KST-based architectures like ActNet suffer from spectral bias [6]. Like standard MLPs, they struggle to fit high-frequency details, making them ill-suited for learning representations of complex images and audio signals. This presents a fundamental gap: can the powerful function decomposition of KST be combined with the high-fidelity representation capabilities of architectures used for learning INRs?

In this paper, we answer this question by introducing **Scaled-ActNet**. Our work is founded on the insight that the spectral bias in KST-based networks is a consequence of their use of fixed, low-frequency basis functions. Scaled-ActNet directly addresses this by making two crucial modifications to the ActNet architecture, inspired by SIREN. First, we introduce input coordinate scaling via a scalar frequency hyperparameter $\omega_0$ that scales the input coordinates before they are processed in each layer, effectively transforming the network's internal low-frequency bases into high-frequency ones. Second, we design a frequency-aware initialization scheme for the network's learnable weights

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$(\Lambda, \beta)$ whose bounds for hidden layers explicitly depend on $\omega_0$, ensuring that activations and gradients are well-behaved and stable throughout the network.

The contributions of the present work are summarized below:

- We identify that current KST-based architectures like ActNet exhibit a spectral bias, fundamentally limiting their utility for learning INRs.
- We propose Scaled-ActNet, which addresses this limitation by integrating a SIREN-inspired input scaling mechanism and a corresponding frequency-aware initialization scheme into the KST framework.
- We provide an open-source PyTorch implementation to enable broader community use and research in KST-based architectures. Code is available at `https://github.com/rishinair05/Scaled-ActNet`.

## 2 Background and Related Work

The Kolmogorov Superposition Theorem (KST) [12] is a fundamental result in function representation theory that establishes an exact decomposition for continuous multivariate functions. For any continuous multivariate function $f : [0,1]^d \to \mathbb{R}$, where $d \geq 2$ is the input dimension, the theorem states that there exist continuous univariate inner functions $\psi_{qp} : [0,1] \to \mathbb{R}$ for $q \in \{1, \ldots, 2d+1\}$ and $p \in \{1, \ldots, d\}$, and continuous univariate outer functions $g_q : \mathbb{R} \to \mathbb{R}$ for $q \in \{1, \ldots, 2d+1\}$, such that for all $\mathbf{x} = (x_1, \ldots, x_d) \in [0,1]^d$:

$$f(x_1, \ldots, x_d) = \sum_{q=1}^{2d+1} g_q \left( \sum_{p=1}^{d} \psi_{qp}(x_p) \right). \tag{1}$$

The inner functions $\psi_{qp}$ are universal and independent of $f$, while the outer functions $g_q$ depend on the target function $f$. This result shows that any $d$-variate function can be exactly decomposed into a finite sum of $2d+1$ outer functions, each applied to a sum of $d$ inner functions, giving $2d^2 + d$ total inner functions. In essence, the theorem shows that multivariate complexity can be reduced to a composition of univariate operations, with addition as the only operation that mixes input dimensions.

**Laczkovich's Representation** Laczkovich [7] provides an alternative representation, which offers greater practical flexibility by allowing a single shared outer function $g$ across all inner functions. In Laczkovich's variant of KST, it is shown that for any continuous function $f : [0,1]^d \to \mathbb{R}$, there exist a finite set of $m$ continuous univariate inner functions $\{\psi_i : [0,1] \to \mathbb{R}\}_{i=1}^m$ where $(2+\sqrt{2})(2d-1) < m \in \mathbb{N}$, a single continuous univariate outer function $g : \mathbb{R} \to \mathbb{R}$, and strictly positive weights $\lambda_{ij} > 0$ for $i \in \{1, \ldots, m\}$ and $j \in \{1, \ldots, d\}$, such that for all $\mathbf{x} \in [0,1]^d$:

$$f(x_1, \ldots, x_d) = \sum_{i=1}^{m} g \left( \sum_{j=1}^{d} \lambda_{ij} \psi_i(x_j) \right). \tag{2}$$

The key differences from the original KST are: (1) the use of a single shared outer function $g$ instead of $2d+1$ distinct functions $g_q$, (2) the introduction of strictly positive weights $\lambda_{ij} > 0$ to scale the inner function outputs, and (3) a flexible number $m$ of terms in the sum.

**The ActNet Architecture** Recently, Guilhoto and Perdikaris [2] propose a new architecture inspired by Laczkovich's representation, referred to as ActNet. The ActNet architecture defines a map $f_\theta : \mathbb{R}^{d_{in}} \to \mathbb{R}^{d_{out}}$ as a composition of layers:

$$f_\theta = P_{out} \circ \mathcal{A}^{(H)} \circ \mathcal{A}^{(H-1)} \circ \cdots \circ \mathcal{A}^{(1)} \circ P_{in}, \tag{3}$$

where $P_{in} : \mathbb{R}^{d_{in}} \to \mathbb{R}^{d_0}$ and $P_{out} : \mathbb{R}^{d_H} \to \mathbb{R}^{d_{out}}$ are affine projection layers (i.e., linear transformations with learnable weight matrices and bias vectors), and $\mathcal{A}^{(h)} : \mathbb{R}^{d_{h-1}} \to \mathbb{R}^{d_h}$ for $h = 1, 2, \ldots, H$ denote the $H$ ActLayers.

The ActLayer drops the outer function $g$ from Laczkovich's formulation, relying on layer composition to recover expressiveness. The $h$-th ActLayer implements the map:

$$\mathbf{z} \mapsto \left( \sum_{j=1}^{d_{h-1}} \lambda_{kj}^{(h)} \phi_k^{(h)}(z_j) \right)_{k=1}^{d_h}, \tag{4}$$

where the inner functions $\phi_k^{(h)} : \mathbb{R} \to \mathbb{R}$ for $k = 1, \ldots, d_h$ are represented as linear combinations of $N_h$ univariate basis functions $b_n^{(h)} : \mathbb{R} \to \mathbb{R}$ for $n = 1, \ldots, N_h$:

$$\phi_k^{(h)}(z_j) = \sum_{n=1}^{N_h} b_n^{(h)}(z_j)\, c_{nk}^{(h)}, \tag{5}$$

where $c_{nk}^{(h)} \in \mathbb{R}$ are learnable expansion coefficients. The $k$-th output component of the $h$-th layer is therefore:

$$[\mathcal{A}^{(h)}(\mathbf{z})]_k = \sum_{j=1}^{d_{h-1}} \lambda_{kj}^{(h)} \sum_{n=1}^{N_h} b_n^{(h)}(z_j)\, c_{nk}^{(h)}. \tag{6}$$

The $h$-th ActLayer can be expressed in matrix form as:

$$\mathcal{A}^{(h)}(\mathbf{z}) = \left( \Lambda^{(h)} \odot \Phi^{(h)}(\mathbf{z}) \right) \mathbf{1}_{d_{h-1}}, \tag{7}$$

where $\Lambda^{(h)} \in \mathbb{R}^{d_h \times d_{h-1}}$ is the weight matrix with entries $[\Lambda^{(h)}]_{kj} = \lambda_{kj}^{(h)} \in \mathbb{R}$, $\mathbf{1}_{d_{h-1}} \in \mathbb{R}^{d_{h-1}}$ is a vector of ones, and $\odot$ denotes the Hadamard (element-wise) product. Note that while Laczkovich's theorem requires $\lambda_{ij} > 0$, ActNet drops this positivity constraint, allowing the weights to take any real value.

The matrix $\Phi^{(h)}(\mathbf{z}) \in \mathbb{R}^{d_h \times d_{h-1}}$ contains the inner function evaluations with entries $[\Phi^{(h)}(\mathbf{z})]_{kj} = \phi_k^{(h)}(z_j)$, and decomposes as $\Phi^{(h)}(\mathbf{z}) = (C^{(h)})^\top B^{(h)}(\mathbf{z})$. Here, $C^{(h)} \in \mathbb{R}^{N_h \times d_h}$ is the coefficient matrix with entries $[C^{(h)}]_{nk} = c_{nk}^{(h)}$, and $B^{(h)}(\mathbf{z}) \in \mathbb{R}^{N_h \times d_{h-1}}$ is the basis evaluation matrix with entries $[B^{(h)}(\mathbf{z})]_{nj} = b_n^{(h)}(z_j)$. Specifically, ActNet uses sinusoidal basis functions of the form $b_n(t) = \sin(\omega_n t + p_n)$, where $\omega_n$ and $p_n$ are learnable frequency and phase parameters, respectively. However, these bases are initialized at low frequencies, which fundamentally restricts the network's ability to represent high-frequency components present in complex signals.

**INRs and Spectral Bias** INRs parametrize signals using a neural network $f_\theta : \mathbb{R}^d \to \mathbb{R}^c$. Standard MLPs with smooth activations (like ReLU or Tanh) suffer from spectral bias, tending to learn low-frequency components first and struggling with high-frequency details [6]. Tancik et al. [9] addressed this by mapping inputs to a higher-dimensional space using random Fourier features $\gamma(\mathbf{x}) = [\cos(2\pi\mathbf{B}\mathbf{x}), \sin(2\pi\mathbf{B}\mathbf{x})]^T$, where the scale of $\mathbf{B}$ controls the representable frequency spectrum. Scaled-ActNet shares a conceptual similarity with Fourier feature networks in that it injects high-frequency components into the model. However, unlike Fourier features which use random projections, Scaled-ActNet scales inputs to explicitly defined KST basis functions.

Other approaches to frequency-adaptive representations include Multiplicative Filter Networks (MFNs) [10] and WIRE [11], which use Gabor wavelets to capture complex signals. SIREN [1] uses a periodic sine activation function, $\sin(\omega_0\mathbf{x})$, to enable the network to model fine details. Our work adapts the core principle of SIREN—frequency scaling via $\omega_0$—to the structured KST framework.

## 3 Scaled-ActNet: A Frequency-Aware Architecture

As discussed earlier, ActNet uses sinusoidal basis functions $b_n(t) = \sin(\omega_n t + p_n)$ with learnable frequencies $\omega_n$ and phases $p_n$ that are initialized at low frequencies. If the target signal contains frequencies not captured by these bases, the network will fail to represent them. This fundamental limitation motivates Scaled-ActNet.

Scaled-ActNet addresses this limitation by scaling the input coordinates before they are passed to the basis functions. Rather than modifying the basis functions themselves, we introduce a frequency

hyperparameter $\omega_0 > 0$ that scales the input. The $h$-th Scaled-ActNet layer $\mathcal{R}^{(h)} : \mathbb{R}^{d_{h-1}} \to \mathbb{R}^{d_h}$ takes an input vector $\mathbf{z} \in \mathbb{R}^{d_{h-1}}$ and produces an output in $\mathbb{R}^{d_h}$. Its $k$-th output component, for $k = 1, \ldots, d_h$, is:

$$[\mathcal{R}^{(h)}(\mathbf{z})]_k = \sum_{j=1}^{d_{h-1}} \lambda_{kj}^{(h)} \sum_{n=1}^{N_h} b_n^{(h)}(\omega_0 z_j)\, c_{nk}^{(h)}, \tag{8}$$

where $\lambda_{kj}^{(h)} \in \mathbb{R}$ are real-valued weights, $b_n^{(h)} : \mathbb{R} \to \mathbb{R}$ for $n = 1, \ldots, N_h$ are the univariate basis functions, and $c_{nk}^{(h)} \in \mathbb{R}$ are the expansion coefficients, all as defined in Section 2. The only modification from the ActLayer (Equation 6) is the scaling of each input component $z_j$ by $\omega_0$.

To see how this scaling modifies the basis functions, note that evaluating $b_n(t) = \sin(\omega_n t + p_n)$ at the scaled input $\omega_0 t$ yields $\sin(\omega_0 \omega_n t + p_n)$, which has effective frequency $\omega'_n = \omega_0 \omega_n$. Since ActNet initialises frequencies as $\omega_n \sim \mathcal{N}(0, 1)$, the $\omega_0$ scaling shifts the initial frequency distribution to $\omega'_n \sim \mathcal{N}(0, \omega_0^2)$, broadening the spectral support of the basis by a factor of $\omega_0$. We note that ActNet also applies a global $\omega_0$ to the input before the first projection layer [2]. In contrast, Scaled-ActNet applies $\omega_0$ scaling within each layer, directly rescaling the basis function arguments at every hidden layer of the network.

In matrix form, the $h$-th Scaled-ActNet layer is:

$$\mathcal{R}^{(h)}(\mathbf{z}) = \left(\Lambda^{(h)} \odot \Phi^{(h)}(\omega_0 \mathbf{z})\right) \mathbf{1}_{d_{h-1}}, \tag{9}$$

where $\Lambda^{(h)} \in \mathbb{R}^{d_h \times d_{h-1}}$, $\Phi^{(h)}$, and $\mathbf{1}_{d_{h-1}}$ are defined as in Section 2, and $\omega_0 \mathbf{z}$ denotes element-wise scaling of the input vector.

For the first layer, we use $\omega_0$, while for subsequent hidden layers we use a separate hyperparameter $\omega_{\text{hidden}}$. Both $\omega_0$ and $\omega_{\text{hidden}}$ are task-specific hyperparameters that must be tuned based on the frequency content of the target signal. In general, $\omega_0$ should be set proportional to the maximum frequency present in the signal: large enough that the scaled basis functions can represent the highest frequencies in the target, but not so large as to introduce optimization instability. Optionally, $\omega_0$ and $\omega_{\text{hidden}}$ can be made trainable parameters, which we found to yield slight improvements in reconstruction quality.

The full Scaled-ActNet architecture follows the same compositional structure as ActNet:

$$f_\theta = P_{out} \circ \mathcal{R}^{(H)} \circ \mathcal{R}^{(H-1)} \circ \cdots \circ \mathcal{R}^{(1)} \circ P_{in}, \tag{10}$$

where $P_{in}$ and $P_{out}$ are affine projection layers as defined in Equation 3. Additionally, the input coordinates are scaled by $\omega_0$ before the initial projection $P_{in}$.

**SIREN-Inspired Initialization** The choice of initialization is critical for networks with periodic activations. In the SIREN architecture, Sitzmann et al. derive an initialization scheme that preserves the distribution of activations across layers: for hidden layers, weights are drawn from $\mathcal{U}(-\sqrt{6/n}, \sqrt{6/n})$, where $n$ is the fan-in, ensuring that the pre-activation values are approximately standard normal. This in turn ensures that the output of each $\sin(\cdot)$ activation follows an arcsine distribution with variance $1/2$, preventing the signal from vanishing or exploding as it propagates through the network. We adapt this scheme to the Scaled-ActNet layer, where $\mathcal{U}(a, b)$ denotes the uniform distribution on the interval $[a, b]$. For the first layer with input dimension $d_{in}$ and $N$ basis functions, the weight matrices are initialized as:

$$[\Lambda^{(1)}]_{kj} \sim \mathcal{U}\left(-\frac{1}{d_{in}}, \frac{1}{d_{in}}\right) \quad \text{and} \quad [C^{(1)}]_{nk} \sim \mathcal{U}\left(-\frac{1}{N}, \frac{1}{N}\right). \tag{11}$$

For subsequent hidden layers, the initialization must compensate for the frequency scaling by $\omega_{\text{hidden}}$ to maintain the same activation variance:

$$[\Lambda^{(h)}]_{kj} \sim \mathcal{U}\left(-\frac{\sqrt{6/d_{h-1}}}{\omega_{\text{hidden}}}, \frac{\sqrt{6/d_{h-1}}}{\omega_{\text{hidden}}}\right) \quad \text{and} \quad [C^{(h)}]_{nk} \sim \mathcal{U}\left(-\frac{\sqrt{6/N_h}}{\omega_{\text{hidden}}}, \frac{\sqrt{6/N_h}}{\omega_{\text{hidden}}}\right). \tag{12}$$

The $\sqrt{6}$ factor arises from the Xavier/Glorot initialization framework adapted to the sinusoidal setting, ensuring approximately constant activation variance across layers. The $1/\omega_{\text{hidden}}$ factor counteracts the frequency scaling so that the product $\omega_{\text{hidden}} \cdot w$ has the same effective magnitude as an unscaled weight, thereby maintaining unit variance under the arcsine distribution induced by the sinusoidal activations.

# 4 Experiments

We evaluate Scaled-ActNet on image and audio representation tasks. We compare against **ActNet** [2] and **SIREN** [1], testing all models at multiple parameter counts. All models were trained using the Adam optimizer [8]. Details are in Appendix A.

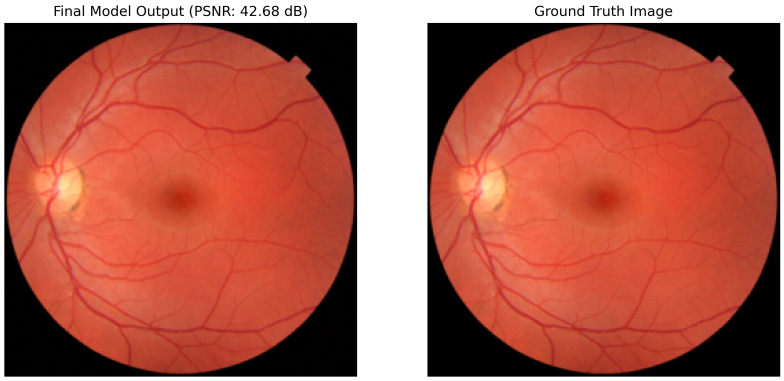
## 4.1 Image Fitting

We test on 256x256 grayscale (Cameraman) and RGB (Retina) images. The network maps 2D coordinates to pixel values. Performance is measured in Peak Signal-to-Noise Ratio (PSNR). All reported values in Table 1 are averaged over 10 independent runs. As shown in the table, Scaled-ActNet outperforms both ActNet and SIREN by up to 23 dB and 15 dB, respectively, on the grayscale Cameraman and RGB Retina test cases.

Table 1: PSNR (dB) on image fitting tasks. Higher is better.

| Model Size | Cameraman (PSNR) | | | Retina (PSNR) | | |
|---|---|---|---|---|---|---|
| | Scaled-ActNet | ActNet | SIREN | Scaled-ActNet | ActNet | SIREN |
| 50k | **34.83** | 24.60 | 26.82 | **35.37** | 26.90 | 30.28 |
| 100k | **43.45** | 24.90 | 28.48 | **41.61** | 27.78 | 31.80 |
| 150k | **44.76** | ∼21.7 | 29.43 | **41.11** | XX.XX | 32.79 |
| 200k | **42.98** | ∼21.7 | 30.02 | **39.70** | XX.XX | 33.43 |

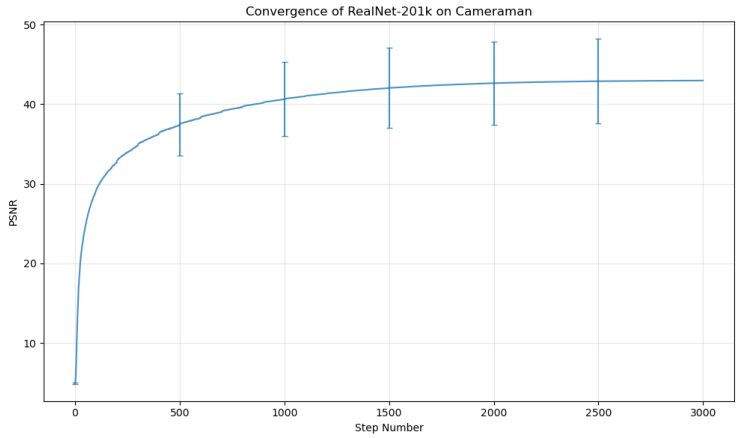
(a) Cameraman Reconstruction (PSNR: 49.52 dB)



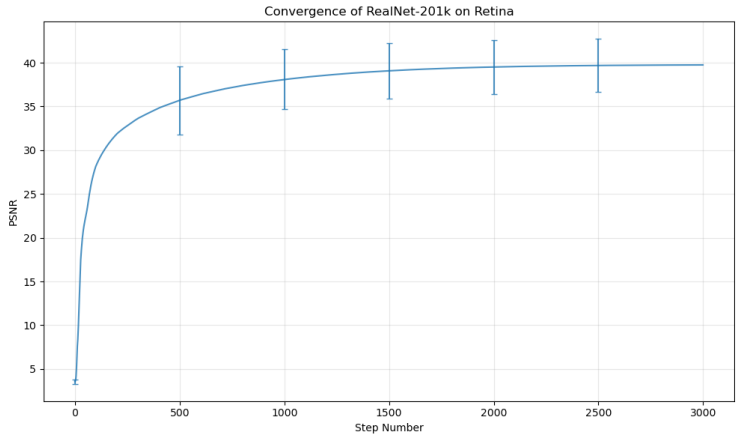

(b) Retina Reconstruction (PSNR: 42.68 dB)

Figure 1: Qualitative reconstruction results for 200k parameter models. Each panel shows the ground truth (left) and Scaled-ActNet's reconstruction (right). Scaled-ActNet achieves exceptionally high fidelity, capturing fine textures and sharp edges.

Figure 1 provides a visual comparison of the best-performing 200k parameter models, showing the single best reconstruction (highest PSNR) out of the 10 runs; Scaled-ActNet's reconstructions are virtually indistinguishable from the ground truth. The convergence plots in Figure 2, averaged over 10 runs for the 200k parameter models, demonstrate that Scaled-ActNet not only reaches a high PSNR but does so with stable and rapid convergence.




(a) Cameraman Convergence




(b) Retina Convergence

Figure 2: Convergence plots for 200k parameter Scaled-ActNet models, averaged over 10 independent runs. Shaded regions indicate one standard deviation.

### 4.2 Audio Fitting

To demonstrate performance on 1D signals, we conduct audio fitting on a waveform dataset (Bach Cello Suite) and compare against SIREN. Performance is measured by Mean Squared Error (MSE) loss. As shown in Table 2, Scaled-ActNet consistently achieves a lower final loss than SIREN. The 200k parameter model achieves a loss nearly 9x lower than SIREN's.

Table 2: Final MSE Loss on the Bach audio fitting task. Lower is better.

| Model Size | Scaled-ActNet | SIREN |
|---|---|---|
| ∼50k | **4.90e-3** | X.Xe-X |
| ∼100k | **3.52e-5** | X.Xe-X |
| ∼150k | **5.84e-5** | X.Xe-X |
| ∼200k | **3.73e-5** | 3.3e-4 |

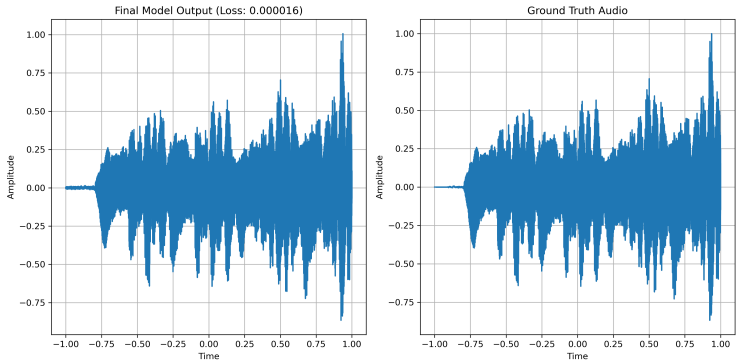



Figure 3: Qualitative reconstruction of the Bach cello waveform for a 200k parameter model. The ground truth and Scaled-ActNet's reconstruction audio waves are presented side by side, showing a near-perfect match. The final MSE loss was 1.6e-5.

## 5 Discussion

**Justification for Task Selection** Our experimental evaluation focuses exclusively on image and audio fitting—canonical tasks for implicit neural representations that require learning a broad and unknown frequency spectrum. We exclude the PDE benchmarks used in the ActNet study [2] due to a methodological concern regarding their experimental initialization. In solving the Poisson equation $\Delta u = f$, the authors employ a ground-truth solution $u(x, y) = \sin(w\pi x)\sin(w\pi y)$ with a varying frequency parameter $w$ from $[2, 32]$. An examination of their implementation reveals that the network's frequency parameter, $\omega_0$, is explicitly initialized to match the parameter of the ground truth equation ($\omega_0 = \pi w$). This practice encodes specific knowledge of the solution's spectral properties directly into the model's initialization. Consequently, the optimization problem is simplified from learning the underlying signal structure to estimating its amplitude. This same initialization flaw applies to the other PDE experiments in [2], including the Allen-Cahn and Helmholtz equations. Such an initialization strategy constitutes a strong inductive bias that is not applicable in general scientific modeling contexts where the true solution's frequency content is unknown. Scaled-ActNet is therefore evaluated on complex, natural signals (images and audio) where the frequency spectrum must be learned from the data itself, ensuring a robust assessment of representational capacity.

**Non-Monotonic Scaling** In our image fitting experiments (Table 1), we observed that for the Cameraman dataset, the PSNR slightly decreases when scaling from 150k to 200k parameters (44.76 dB to 42.98 dB). This non-monotonic behavior likely indicates optimization difficulties at larger model sizes, suggesting that simple scale-up strategies may yield diminishing returns without further regularization or hyperparameter tuning.

**Computational Considerations** While Scaled-ActNet significantly outperforms ActNet in reconstruction quality, it introduces additional computational overhead due to the input scaling and slightly more complex initialization logic. However, during inference, the cost is identical to ActNet, as the scaling is a simple element-wise multiplication. Compared to SIREN, Scaled-ActNet requires calculating the fixed basis expansion, which can be memory-intensive for very large basis sets, but offers potentially greater expressivity per parameter for certain classes of functions.

## 6 Conclusion

We have introduced Scaled-ActNet, a neural architecture that resolves the critical issue of spectral bias in KST-based networks. By incorporating a SIREN-inspired input scaling parameter, $\omega_0$, and a corresponding principled initialization scheme, Scaled-ActNet excels at high-frequency signal representation where previous models like ActNet struggle. Our experiments in image and audio representation learning tasks show dramatic improvements over baselines, including up to 23 dB PSNR in image fitting and nearly 9x lower loss in audio fitting. This work demonstrates that the theoretical strengths of KST and the practical advantages of frequency-aware INRs are not mutually exclusive, paving the way for a new class of versatile, high-fidelity neural networks.

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## A Implementation Details

All models were trained on a single NVIDIA 4090 GPU. The architecture for all experiments consists of an initial linear layer, 3 hidden Scaled-ActNet layers, and a final linear output layer.

### A.1 Image Fitting Task

- **Datasets:** Cameraman (256x256, Grayscale) and Retina (256x256, RGB).
- **Shared Hyperparameters:**
  - Optimizer: Adam, with a learning rate scheduler that decays the learning rate by a factor of 0.8 every 100 epochs.
  - Number of Epochs: 3,000
  - Batch Size: Full image ($256 \times 256$ = 65,536 coordinates)
  - First Layer Frequency $\omega_0$: 15.0
  - Hidden Layer Frequency $\omega_{\text{hidden}}$: 30.0
- **Per-Size Hyperparameters:**

| Model Size (Params) | Embedding/Hidden Dim | Learning Rate |
|---|---|---|
| $\sim$50k | 128 | 1e-3 |
| $\sim$100k | 180 | 5e-4 |
| $\sim$150k | 221 | 1e-4 |
| $\sim$200k | 256 | 1e-4 |

### A.2 Audio Fitting Task

- **Dataset:** Bach cello clip (7.0s at 44.1kHz).
- **Shared Hyperparameters:**
  - Optimizer: Adam, with a learning rate scheduler that decays the learning rate by a factor of 0.8 every 100 epochs.
  - Number of Epochs: 3,000
  - Batch Size: Full clip
  - First Layer Frequency $\omega_0$: 80.0
  - Hidden Layer Frequency $\omega_{\text{hidden}}$: 80.0
- **Per-Size Hyperparameters:** The same dimensions and learning rates per model size were used as in the image fitting task.